Practice workbook · Explain how temperature or pH changes enzyme activity

Explaining rising, falling or zero enzyme activity as temperature or pH changes, why tests are kept at 37 °C or one pH, and why body temperature and pH must stay near normal.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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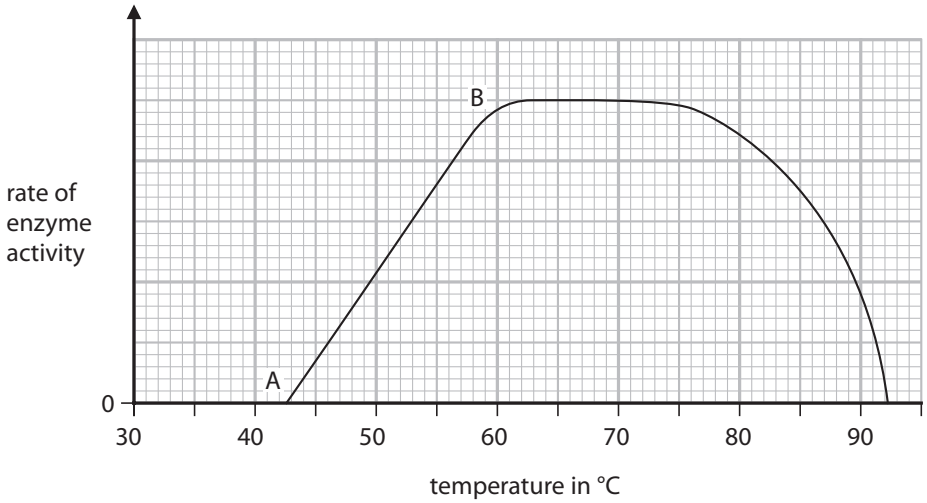
Type ID: HBI-ENZYME-TEMP-PH-DENATURE

1. Question 4 · part b · 5 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b only.

4 The graph shows the effect of temperature on the activity of a bacterial enzyme.



(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

(2)

(b) Explain what is causing the change in enzyme activity from A to B.

(5)

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1. Question 4 · part b · 5 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b only.

(c) Deduce which type of environment the bacterium lives in.

(2)

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(Total for Question 4 = 9 marks)

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Working and answer

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2. Question 3 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

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3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

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2. Question 3 · part b.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

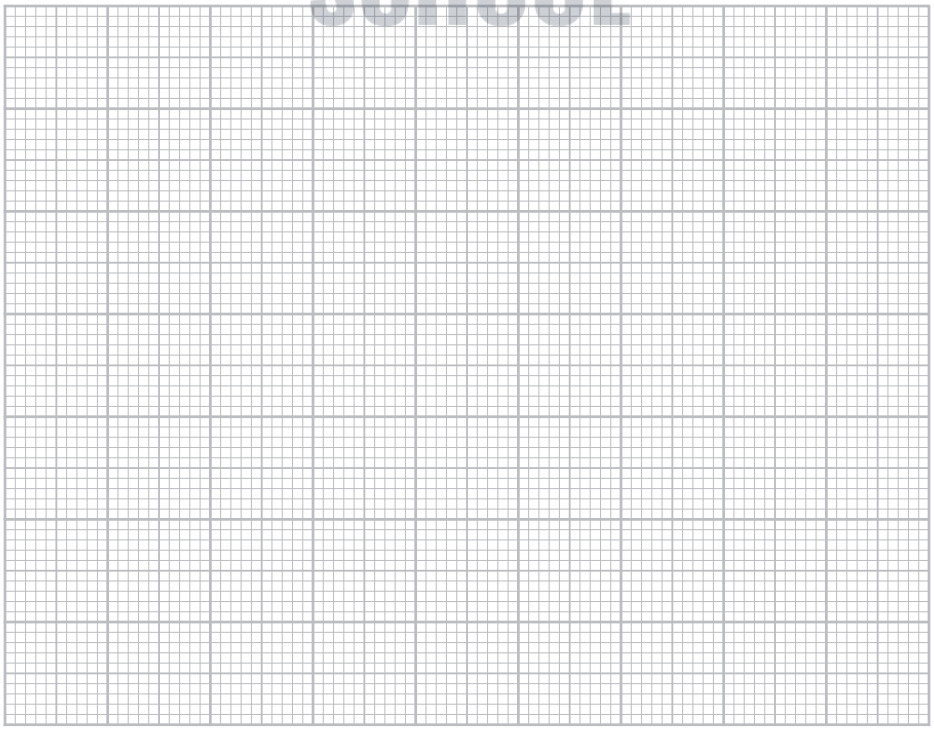
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



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P 6 6 4 3 2 A 0 8 2 4

2. Question 3 · part b.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

(3)

(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

(3)

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(Total for Question 3 = 15 marks)

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2. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

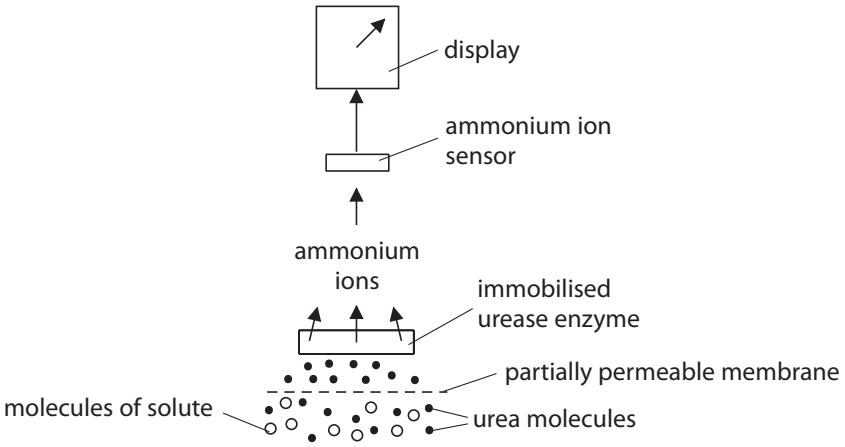
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3. Question 3 · part b · 4 marks

June 2021 Paper 2

Further practice · Attempt part b only.

3 The diagram shows a biosensor that can be used to measure the concentration of urea in urine.



(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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3. Question 3 · part b · 4 marks

June 2021 Paper 2

Further practice · Attempt part b only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

(4)

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

3. Question 3 · part b · 4 marks · Working and answer

Working and answer

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4. Question 4 · part b.iii · 4 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

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4. Question 4 · part b.iii · 4 marks

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Further practice · Attempt part b.iii only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

(1)

(ii) Give the independent variable and the dependent variable for this investigation.

(2)

independent

dependent

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(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

(4)

(Total for Question 4 = 13 marks)

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4. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

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5. Question 5 · part b · 4 marks

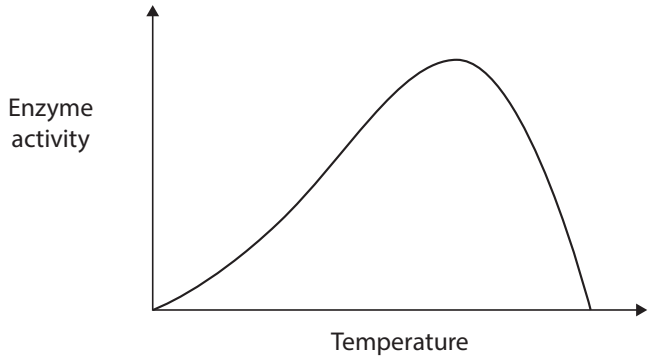
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Further practice · Attempt part b only.

5 (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.



(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

(5)

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5. Question 5 · part b · 4 marks

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Further practice · Attempt part b only.

(ii) State the independent variable in this investigation.

(1)

.....

(b) Explain how enzyme activity is affected by low temperatures.

(4)

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(Total for Question 5 = 10 marks)

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5. Question 5 · part b · 4 marks · Working and answer

Working and answer

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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

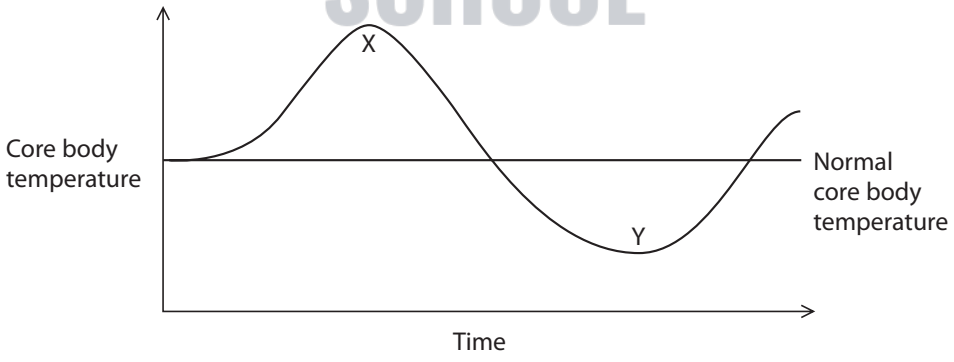
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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6. Question 8 · part c · 5 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

(c) Hypothermia is a condition where the core body temperature drops below 35 °C.

Explain why it is important to increase the core body temperature of people with hypothermia to within the normal range.

(5)

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(Total for Question 8 = 14 marks)

6. Question 8 · part c · 5 marks · Working and answer

Working and answer

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7. Question 7 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

7 Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

Student	Volume of enzyme solution in cm^3	Temperature in °C	pH	Volume of cloudy suspension in cm^3	Time tubes left in minutes	Appearance at end of investigation
1	5	37	2	10	9	cloudy
2	5	37	2	10	20	clear
3	5	37	4	10	35	clear
4	5	37	4	10	20	cloudy
5	5	37	6	10	8	cloudy
6	5	37	6	10	40	cloudy

(a) Explain why the investigation was carried out at 37 °C.

(2)

(b) Explain which property of the enzyme the students are investigating.

(2)

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7. Question 7 · part a · 2 marks · Working and answer

Working and answer

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8. Question 3 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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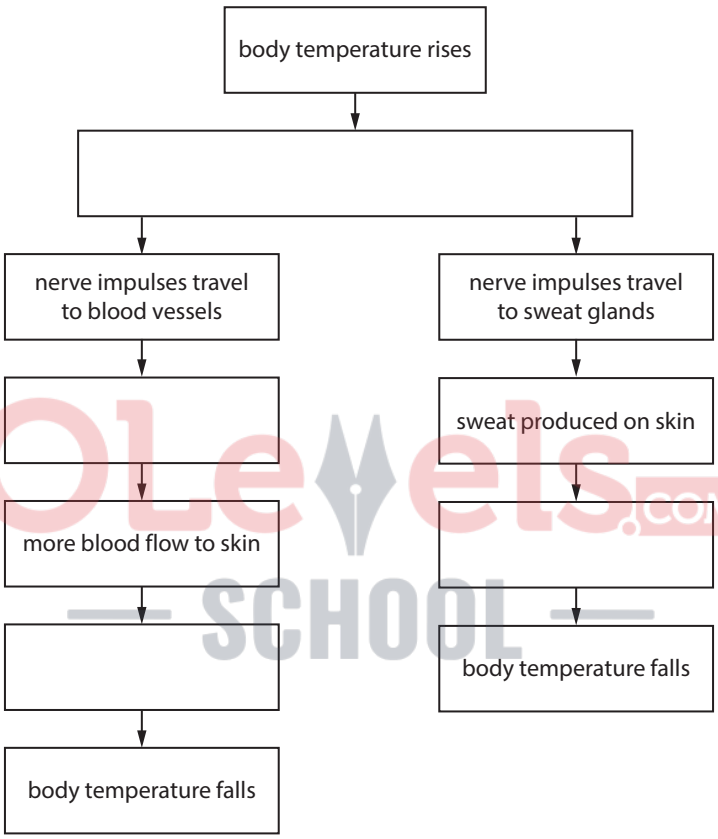
8. Question 3 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

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(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

P 7 8 7 6 8 A 0 7 2 0

Turn over ▶

8. Question 3 · part c · 4 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(c) Explain why it is important that body temperature should not become too high.

(4)

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(Total for Question 3 = 14 marks)

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Working and answer

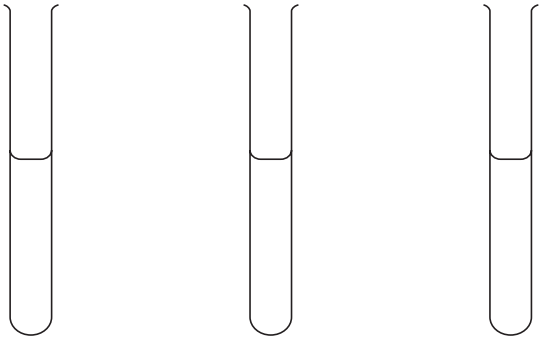
9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

2 A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.



A **B** **C**

Table 1 lists the contents of each test tube.

Test tube A	Test tube B	Test tube C
5 cm^3 of milk	5 cm^3 of milk	5 cm^3 of milk
2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate
6 drops of phenolphthalein	6 drops of phenolphthalein	6 drops of phenolphthalein
1 cm^3 enzyme	1 cm^3 enzyme	1 cm^3 boiled enzyme
distilled water	bile salts	bile salts

Table 1

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9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

	Colour of contents		
	Tube A	**Tube B**	**Tube C**
at start	pink	pink	pink
after 5 minutes	pink	colourless	pink
after 10 minutes	pink	colourless	pink
after 15 minutes	colourless	colourless	pink

Table 2

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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9. Question 2 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(b) Explain the purpose of tube C.

(2)

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

(2)

(d) Describe the effect of bile salts on lipid digestion in this investigation.

(3)

(Total for Question 2 = 10 marks)

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9. Question 2 · part c · 2 marks · Working and answer

Working and answer

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10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

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(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

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(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

10. Question 1 · part b.iii · 4 marks

November 2021 Paper 1

Further practice · Attempt part b.iii only.

(b) (i) State the condition where the enzyme was most active.

(1)

(ii) State the reason for including the beaker with no hydrogen peroxide added.

(1)

(iii) Explain the result obtained when the conditions were acidic.

(4)

(Total for Question 1 = 14 marks)

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10. Question 1 · part b.iii · 4 marks · Working and answer

Working and answer

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11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

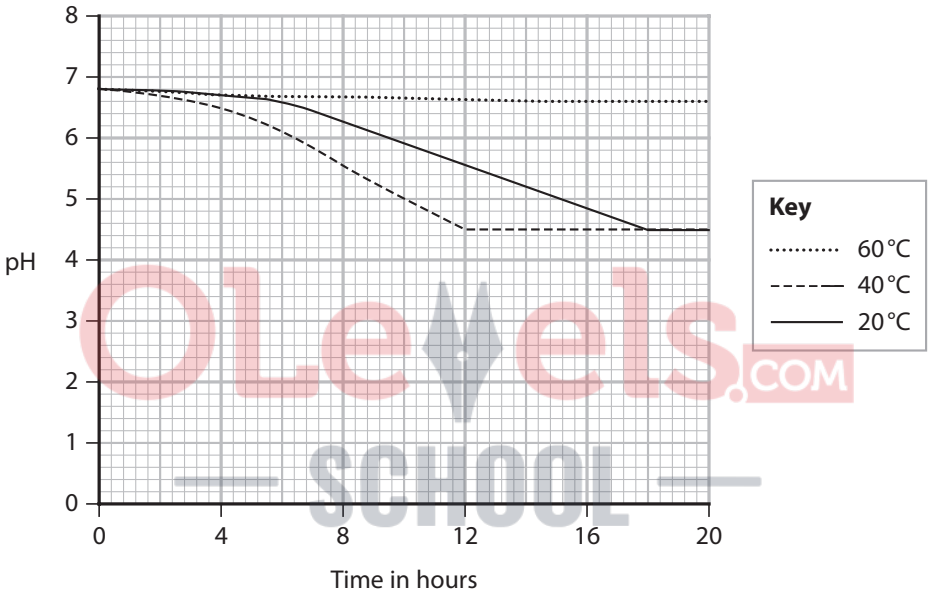
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

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(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

.....

.....

cooling

.....

.....

11. Question 6 · part b · 2 marks

January 2023 Paper 1

Further practice · Attempt part b only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

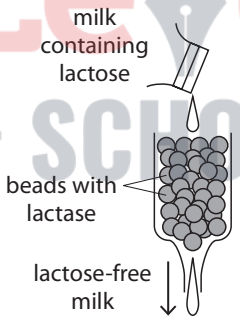
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

(2)

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.



(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

(2)

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P 7 2 2 7 1 A 0 1 8 2 4

11. Question 6 · part b · 2 marks · Working and answer

Working and answer

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12. Question 2 · part b.ii · 2 marks

June 2023 Paper 2R

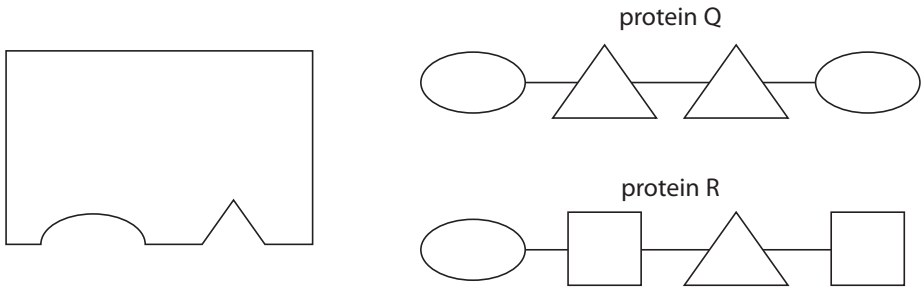
Further practice · Attempt part b.ii only.

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2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases
protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

.....

12. Question 2 · part b.ii · 2 marks

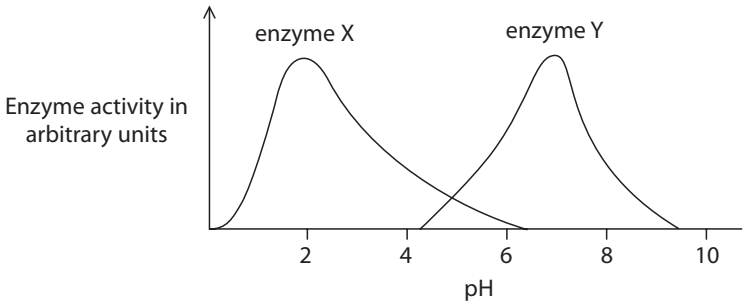
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Further practice · Attempt part b.ii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

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P 7 2 5 8 8 A 0 6 2 4

12. Question 2 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(iii) Describe how the student could make the results of this investigation more reliable.

(2)

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(iv) Name one control variable in the student's investigation.

(1)

.....

(Total for Question 2 = 13 marks)

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Working and answer

13. Question 4 · part a.v · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.v only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

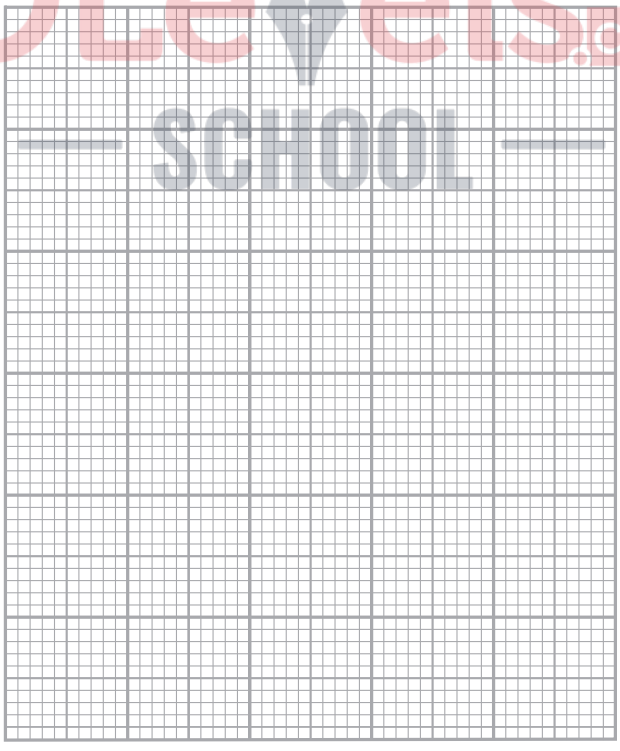
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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13. Question 4 · part a.v · 3 marks

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Further practice · Attempt part a.v only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

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(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

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(Total for Question 4 = 13 marks)

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13. Question 4 · part a.v · 3 marks · Working and answer

Working and answer

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14. Question 4 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

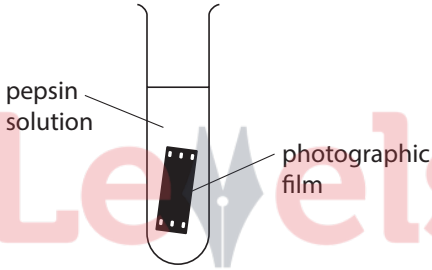
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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14. Question 4 · part b.ii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part b.ii only.

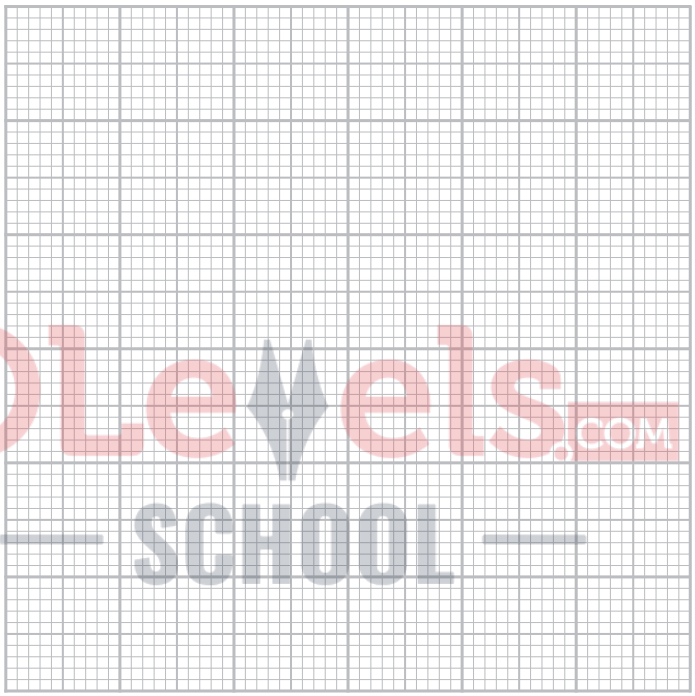
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



(b) (i) Explain why the photographic film becomes clear.

(3)

14. Question 4 · part b.ii · 2 marks

June 2025 Paper 1R

Further practice · Attempt part b.ii only.

(ii) Explain which would be the best pH to carry out this investigation.

(2)

(iii) State three variables that should be controlled during this investigation other than pH.

(3)

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

(2)

(Total for Question 4 = 15 marks)

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14. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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